# Ahmad Mustafa Anis

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## Professional Summary

AI/ML Research Engineer with 4+ years of industry experience building and deploying deep learning systems across computer vision, video understanding, and natural language processing. Published at CVPR, ICLR, and NeurIPS; currently conducting research in Self-Supervised Learning at Carnegie Mellon University's TarrLab. Experienced across the full research-to-production pipeline, from model architecture and fine-tuning to real-time inference optimization and multimodal system design.

## Education

**Carnegie Mellon University** — Pittsburgh, PA
*Master of Science in Artificial Intelligence Engineering.* — *Dec 2027*

**International Islamic University** — Islamabad, Pakistan
*Bachelor of Science in Computer Science.* — *Sep 2022*

## Experience

**Research Assistant** — Jun 2026 – Present
*TarrLab, Carnegie Mellon University*

- Investigate progressive learning strategies to improve Self-Supervised Learning, supervised by Professor Michael Tarr and Professor Deva Ramanan.

**AI Research Fellow** — Sep 2024 – Jun 2025
*Fatima Fellowship*

- Researched Retrieval-Augmented Test Time Inference for Medical Vision Language Models, advised by Dr. Wei Peng from Stanford University.

**Deep Learning Engineer** — Mar 2024 – Aug 2026
*Roll AI*

- Optimized real-time video bokeh effects by implementing Depth Estimation and Matting algorithms, achieving 2x performance improvement through parallel processing and batch optimization for live streaming applications.
- Developed and deployed an Active Speaker Detection system for multi-camera live events, automating speaker-camera associations and reducing event video editing time from 48 hours to 1 hour (98% improvement).
- Built a Video Understanding pipeline for automated chapter generation and captioning, enabling intelligent b-roll suggestions by analyzing semantic content and scene transitions across long-form video.
- Developed a multi-view video generation pipeline that synthesizes novel viewpoints from temporal endpoints, enabling enhanced scene reconstruction for live event production.
- Architected a multimodal LLM Agent as a centralized AI video editing system, automating social media clip generation, multi-take video orientation and cropping, transcript correction and enhancement, and dynamic VFX and motion graphics creation to streamline post-production workflows.

**AI Fellow** — Dec 2023 – Feb 2024
*PI School of AI*

- Selected as 1 of 10 fellows from 300 global applicants for a fellowship valued at 12,500 EUR; extracted key information from long-context interview transcripts using LLMs (MistralLite, Mixtral8x7B) and built RAG pipelines using DSPy.

**Machine Learning Engineer** — Apr 2022 – Mar 2024
*Red Buffer*

- Built a fact verification and identification pipeline with multi-threading for a 200% speed increase; fine-tuned LLaMAv2, Dolly-3B, GPT-Neo, and BART for negation generation to detect false news.
- Reduced OCR latency from 0.6 FPS to 16 FPS on CPU for streaming devices to blur sensitive information.
- Boosted logo recognition accuracy from 30% to 85% during cricket matches using CLIP by OpenAI.
- Developed a drag-and-drop RAG system compatible with any LLM, vector DB, and multi-hop search.

**Deep Learning and Computer Vision Intern** — Jul 2021 – Dec 2021
*WortelAI*

- Developed a Safety Clothes Detection system for factory workers using YOLOv5.
- Trained a Celebrity Detection and Classification system using 3DDFAv2, ResNet50, and a 172GB dataset.
- Built a real-time video search engine using CLIP by OpenAI, Qdrant for vector DB, and ElasticSearch.
- Classified multi-label text using BERT and PyTorch across 1,500+ labels.

## Projects

**DistAYA**

- Contributed to the AYA Expedition project focused on creating small multilingual LLMs distilled from large multilingual LLMs, published on Hugging Face.

## PUBLICATIONS

- A. M. Anis, Hasnain Ali, M. Saquib Sarfraz. On the Limitations of Vision-Language Models in Understanding Image Transforms. CVPR 2025 Workshop: Computer Vision in the Wild.
- S. Longpre, N. Singh, ... A. M. Anis (11th Author), et al. Bridging the Data Provenance Gap Across Text, Speech, and Video. ICLR 2025.
- S. Longpre, R. Mahari, ... A. M. Anis (16th Author), et al. Consent in Crisis: The Rapid Decline of the AI Data Commons. NeurIPS 2024.

## OPEN SOURCE CONTRIBUTIONS

- Pandas (pandas-dev/pandas): Contributed bug fixes and improvements to the core data manipulation library.
- LlamaIndex (run-llama/llama_index): Contributed to the open-source LLM application framework.
- Google Research BigVision (google-research/big_vision): Contributed to the large-scale vision research codebase.
- OpenAI Shap-E (openai/shap-e): Contributed to the text-to-3D generation model codebase.

## INTERVIEWS, PODCASTS, AND TECHNICAL WRITING

- Interview with AtomCamp: Provided advice for beginners in Machine Learning.
- Podcast, Adventures in Machine Learning: Discussed optimizing and improving OCR systems.
- Podcast, Adventures in Machine Learning: Discussed deploying Deep Learning models with FastAPI.
- Technical writing published on Medium, KDNuggets, and CNVRG.

## ACTIVITIES & LEADERSHIP

**Data Provenance Initiative, MIT Media Lab** — Dec 2023 - Present
*Research Contributor*
- Audit LLM training data to assess web data access shifts, cultural representation, and transparency.

**Cohere Labs Community** — Nov 2022 - Present
*Volunteer Researcher and Regional Community Lead*
- Organized 50+ guest speaker sessions with Asian researchers; initiated NYU Deep Learning study group (Prof. Yann LeCun); served as Urdu Language Lead for AYA and contributed 3 Urdu datasets.

**MESA Program at Foothill College** — Jan 2024 - Present
*Mentor*
- Provide mentorship to underrepresented students in STEM.

**Bytewise Limited** — Jun 2024 - Aug 2024
*Fellowship Lead (NLP)*
- Mentored 20 students and supervised the NLP curriculum.

**Amazon Web Services** — Aug 2022 - Apr 2023
*AWS Community Builder*
- Organized community events to raise awareness about AWS services.

## SKILLS

**Libraries and Frameworks:** PyTorch, PyTorch Lightning, Distributed Training, TensorFlow, OpenCV, FastAPI, Flask

**Programming Languages:** Python, C++, Rust, Java, JavaScript

**Databases:** MySQL, PostgreSQL

**Tools:** Git, GitHub, GitLab, Docker, Jupyter Notebook, Anaconda, Visual Studio Code

## CERTIFICATIONS

- Microsoft Technology Associate (MTA) 98-381: Introduction to Programming Using Python
- Machine Learning - Coursera
- Deep Learning Specialization - Coursera
- Getting Started with TensorFlow 2 - Coursera
- NYU Deep Learning by Yann LeCun
- Mathematics for Machine Learning - Imperial College London

## AWARDS & HONORS

- 2,600+ points on StackOverflow, impacting 300,000+ people
- 5 kyu, top 36% worldwide on Codewars
- Platinum badge on KDNuggets